

2 Asset Pricing

Lucas (1978) (aka “Lucas-tree model”)¹ studies the determination of equilibrium asset prices in a one-good, pure exchange economy with identical consumers.

Trees and Assets

- Single consumption good (apples), nonstorable
- There are k productive units (trees) in fixed supply
- Trees (costlessly) grow apples
- Productivity of each tree fluctuates stochastically over time.

An **asset** is a claim to all or part of the output of one of these units (claim to a stochastic dividend stream).

Since productivities fluctuate stochastically over time, so will the equilibrium asset prices. Our goal is to understand the relationship between these exogenously determined productivities changes and market-determined movements in asset prices.

Consumers and Preferences

There is a continuum of measure 1 of identical consumers (same preferences and equal endowments of assets). Alternatively, we can think of one representative consumer.

Preferences are given by

$$E \sum_{t=0}^{\infty} \beta^t U(c_t),$$

where c_t is a stochastic process for consumption and $U: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is bounded, continuously differentiable, strictly increasing, strictly concave, with $U(0) = 0$.

(Exogenous) Production

- Let $Z \subseteq \mathbb{R}_+^k$ be a compact subset of $\mathbb{R}_+^k$ with its Borel subsets $\mathcal{Z}$

¹See also SLP Chapter 10.6

- z_{it} is the output of unit i in period t
- $z_t = (z_{1t}, z_{2t}, \dots, z_{kt}) \in Z$ is the output vector in period t
- Feasible consumption levels:

$$0 \leq c_t \leq \sum_{i=1}^k z_{it}$$

- z_t follows a Markov process, with a stationary transition function Q on $(Z, \mathcal{Z})$
- Q has the Feller property.

Markets

In each period there are spot markets for the single consumption good and for shares in the productive units.

Each unit has one perfectly divisible equity share. A share entitles its owner, as of the beginning of period t , to all of the unit's output in period t .

Prices

Our goal is to characterize asset prices in a *stationary* competitive equilibrium.

Normalize the price of the current consumption good to one. In any stationary equilibrium, asset prices in each period are described by a *stationary* function $p: Z \rightarrow \mathbb{R}_+^k$, so that $p(z) = (p_1(z), p_2(z), \dots, p_k(z))$ is the vector of asset prices in state $z \in Z$.

The Consumer's Problem in a Sequence Form

Denote the representative consumer's beginning-of-period- t asset holdings by $x_t = (x_{1t}, x_{2t}, \dots, x_{kt})$. The consumer chooses consumption and the next period's asset holdings to maximize his expected utility

$$\max_{\{c_t(z^t), x_{t+1}(z^t)\}_{t=0}^{\infty}} E \sum_{t=0}^{\infty} \beta^t U(c_t(z^t))$$

subject to the following budget constraints:

$$\begin{aligned} c_t(z^t) + p(z_t)x_{t+1}(z^t) &\leq [z_t + p(z_t)]x_t(z^{t-1}) \quad \forall z^t, \forall t \\ c_t(z^t), x_{t+1}(z^t) &\geq 0 \quad \forall z^t, \forall t \end{aligned}$$

The consumer's decision rules are

$$\begin{aligned}c_t &= c(x_t, z_t, p_t), \\x_{t+1} &= g(x_t, z_t, p_t),\end{aligned}$$

where $p_t = p(z_t)$.

In this economy all output will be consumed ($c_t = \sum_i z_{it}$) and all shares will be held ($x_t = (1, 1, \dots, 1) \equiv \underline{1} \forall t$). The main analytical issue, then, will be to determine prices that support this allocation as a competitive equilibrium

Recursive Problem

We now want to use recursive techniques to formulate the consumer's problem.

First, it will be convenient to restrict the space for x . Since we know that in equilibrium the consumer chooses to hold exactly one unit of each asset, any bound $\bar{x} > 1$ is non-binding in equilibrium.

Let $X = [0, \bar{x}]^k$, with its Borel subsets $\mathcal{X}$. Assume for simplicity that $Z \subset \mathbb{R}_{++}^k \Rightarrow$ strictly positive price function.

Define $\Gamma: X \times Z \rightarrow X$ by

$$\Gamma(x, z) = \{y \in X : p(z)y \leq (z + p(z))x\}.$$

Bellman Equation:

$$v(x, z) = \max_{y \in \Gamma(x, z)} U[zx + p(z)(x - y)] + \beta \int v(y, z') Q(z, dz').$$

We can use our stochastic DP techniques to show that for any continuous, strictly positive function $p: Z \rightarrow \mathbb{R}_{++}^k$, there exists a unique $v: X \times Z \rightarrow \mathbb{R}_+$ that solves the problem above. For any $z \in Z$, v is strictly increasing and *weakly* concave in x . The associated policy correspondence G is nonempty, compact-valued, convex-valued and u.h.c.

Exercise. *Why is v only weakly concave? Is it a problem here that we do not have strict concavity of v ?*

Fix $(x, z) \in X \times Z$. We can also show that if $x \in \text{int}X$ and $G(x, z) \cap \text{int}\Gamma(x, z)$ is nonempty, then $v(\cdot, z)$ is differentiable with respect to (w.r.t.) x at (x, z) . Thus, we

can take F.O.C. and Envelope Condition. Denote $v_i(x, z) \equiv \frac{\partial v}{\partial x_i}(x, z)$.

- FOC y_i : $U'[zx + p(z)(x - y)]p_i(z) = \beta \int v_i(y, z')Q(z, dz')$
- Env. x_i : $v_i(x, z) = U'[zx + p(z)(x - y)](z_i + p_i(z))$

for $i = 1, \dots, k$, $y \in G(x, z)$.

Now we will use the fact that in equilibrium p must be such that each consumer starting with $x = \underline{1}$ chooses $y = \underline{1}$. Note that since $z \in \mathbb{R}_{++}^k$, it follows that $z \cdot \underline{1} > 0$, $\forall z \in Z \implies \underline{1} \in \text{int } \Gamma(\underline{1}, z) \forall z \in Z$.

Equilibrium Price Function

The equilibrium price function, which we denote by p^* (and the corresponding value function by v^*) satisfies:

- Env.: $v_i^*(\underline{1}, z) = U'(z \cdot \underline{1})(z_i + p_i^*(z))$
- FOC: $U'(z \cdot \underline{1})p_i^*(z) = \beta \int v_i^*(\underline{1}, z')Q(z, dz')$

for $i = 1, \dots, k$.

Combining both equations, we get the *stochastic Euler equation*

$$U' \left(\sum_j z_j \right) p_i^*(z) = \beta \int U' \left(\sum_j z'_j \right) [z'_i + p_i^*(z')] Q(z, dz'), \quad (1)$$

Remark (Stochastic Discount Factor and Consumption CAPM) *Related to the above stochastic Euler equation is the concept of a stochastic discount factor, or a pricing kernel, widely used in Asset Pricing. Rewrite (1) as*

$$p_{i,t}^* = E_t \left(\beta \frac{U'(c_{t+1})}{U'(c_t)} [z_{i,t+1} + p_{i,t+1}^*] \right) = E_t \sum_{s=t+1}^{\infty} \left(\underbrace{\beta^{s-t} \frac{U'(c_s)}{U'(c_t)}}_{=m_s} z_{i,s} \right).$$

Then $\{m_s\}$ is a stochastic discount factor. Notice that if an asset pays a dividend in states when c_s is high and so the marginal utility is low, that dividend will be discounted by more, and thus have a smaller effect on price, than if it paid in a state when the marginal utility is high.

Consider, for simplicity, a two-period example, where an asset (traded in the first period) only pays a dividend in the second period. Then the expression for the price becomes

$$p_t = E_t(m_{t+1}z_{t+1}),$$

where $m_{t+1} = \beta U'(c_{t+1})/U(c_t)$. Alternatively, rewriting this equation in terms of the gross return $R_{t+1} = z_{t+1}/p_t$, we have

$$1 = E_t(m_{t+1}R_{t+1}),$$

or, equivalently,

$$1 = E_t(m_{t+1})E_t(R_{t+1}) + \text{Cov}(m_{t+1}, R_{t+1}).$$

A risk-free asset with constant return R^f has a zero covariance with the discount factor, and thus

$$R^f = \frac{1}{E_t(m_{t+1})}.$$

Inserting this in

$$p_t = E_t(m_{t+1})E_t(z_{t+1}) + \text{Cov}(m_{t+1}, z_{t+1}),$$

we obtain

$$p_t = \frac{E_t(z_{t+1})}{R^f} + \text{Cov}(m_{t+1}, z_{t+1}).$$

The first term is the standard discounted present value formulation, the second term is a risk adjustment. This equation says that prices are equal to the expected cashflow discounted at the risk-free rate, plus a risk premium. The risk premium depends of the covariance of the payoff with the discount factor. This covariance is typically a negative number (most assets pay off well in good times), so most assets have a lower price than otherwise (or a higher average return) as compensation for risk.

Applying the covariance decomposition for expected returns and using $R^f = 1/E_t(m_{t+1})$:

$$\begin{aligned} E_t(m_{t+1})E_t(R_{t+1}) - 1 &= -\text{Cov}(m_{t+1}, R_{t+1}), \\ E_t(R_{t+1}) - R^f &= -R^f \text{Cov}(m_{t+1}, R_{t+1}), \\ E_t(R_{t+1}) - R^f &= -\frac{\text{Cov}(U'(c_{t+1}), R_{t+1})}{E(U'(c_{t+1}))}. \end{aligned}$$

Assets whose returns co-vary positively with consumption make consumption more

volatile and must promise investors higher expected returns to induce investors to hold them. The covariance between an asset's return and consumption is known as its "consumption beta."

Any continuous function p^* satisfying equation (1) is an equilibrium price function and $v^*(x, z) \equiv v(x, z; p^*)$ is the corresponding value function.

Our goal is to show that there exists a unique p^* that solves (1).

Define

$$h_i(z) = \beta \int U' \left(\sum_j z'_j \right) z'_i Q(z, dz')$$

for $i = 1, \dots, k$.

Then finding an equilibrium price function $p^*(z) = [p_1^*(z), \dots, p_k^*(z)]$ is equivalent to finding functions $f_1(z), \dots, f_k(z)$ satisfying the following k (independent) functional equations

$$f_i(z) = h_i(z) + \beta \int f_i(z') Q(z, dz') \quad (2)$$

for $i = 1, \dots, k$.

Then

$$p_i^*(z) = \frac{f_i(z)}{U' \left(\sum_j z_j \right)}$$

for $i = 1, \dots, k$ is a solution to (1).

Define $T_i f_i(z)$ to be the right-hand side of (2). Using the CMT we can show that $\forall i = 1, \dots, k$ there exists a unique bounded continuous function f_i satisfying (2). We can also show that each f_i is strictly positive.

Thus, the paper develops a general method of constructing equilibrium prices.

Finally, we can calculate the expected return of asset i as follows:

$$E_t[R_{t+1}^i | z] = \frac{\int [z'_i + p_i^*(z')] Q(z, dz')}{p_i^*(z)}$$

(reflects how many \$ of earnings you derive from each \$ of the asset you purchased).

Linear Utility

In this case (1) becomes

$$\begin{aligned} p_i^*(z) &= \beta E(z'_i|z) + \beta E(p_i^*(z')|z), \\ p_i^*(z) &= \beta E(z'_i|z) + \beta^2 E(z''_i|z) + \beta^2 E(p_i^*(z'')|z), \\ p_{i,t}^*(z) &= \sum_{s=1}^{\infty} \beta^s E(z_{i,t+s}|z_{it} = z). \end{aligned}$$

That is, the price of the i -th asset is the expected discounted present value of its real dividend stream, conditional on current information z .

One Asset, Independent Shock

Let $\phi(z)$ be the c.d.f of z

$$h(z) = \beta \int U'(z') z' d\phi(z') = \beta E[z' U'(z')] \equiv \bar{h},$$

independent of z .

$$(Tf_i)(z) = h_i(z) + \beta \int f_i(z') Q(z, dz')$$

becomes

$$(Tf)(z) = \bar{h} + \beta \int f(z') d\phi(z').$$

Calculating f as, say, $\lim_{n \rightarrow \infty} T^n 0$, we get

$$f(z) = \frac{\bar{h}}{1 - \beta},$$

independent of z . Hence $f'(z) = 0$.

Recall

$$p(z) = \frac{f(z)}{U'(z)} = \frac{\bar{h}}{1 - \beta} \frac{1}{U'(z)}.$$

Differentiating this expression gives

$$p'(z) = -\frac{\bar{h}}{1 - \beta} \frac{U''(z)}{[U'(z)]^2} = -p(z) \frac{U''(z)}{U'(z)} > 0$$

since $p > 0$ and U is strictly concave.

This implies that

$$\underbrace{z \frac{p'(z)}{p(z)}}_{\substack{\text{elasticity} \\ \text{of price} \\ \text{wrt income}}} = \underbrace{-z \frac{U''(z)}{U'(z)}}_{\substack{\text{Arrow-Pratt} \\ \text{measure of relative} \\ \text{risk aversion}}} > 0$$

This is an income effect. When income increases ($\uparrow z$), agents attempt to distribute part of the windfall over future periods, via securities purchases. This leads to increases in asset prices.

One Asset, Correlated Shock

Differentiating $p(z) = f(z)/U'(z)$,

$$p'(z) = -\frac{f(z)U''(z)}{[U'(z)]^2} + \frac{f'(z)}{U'(z)} = -\frac{p(z)U''(z)}{U'(z)} + \frac{f'(z)}{f(z)}p(z) \Rightarrow$$

$$z \frac{p'(z)}{p(z)} = \underbrace{-z \frac{U''(z)}{U'(z)}}_{\substack{\text{"income effect"} \\ > 0}} + \underbrace{z \frac{f'(z)}{f(z)}}_{\substack{\text{"information effect"} \\ \text{same sign as } f'(z)}}$$

The information effect can be positive or negative. Suppose a higher z today implies that the agents are more optimistic about future dividends. This can cause them to demand more or less of the asset. Basically, there is an income and a substitution effect, and the net effect depends on which of these two effects dominates. (This is a classical Becker's story: a decrease in child mortality (increase in expected return) need not mean that people choose to have more kids. That's the example with $f'(z) < 0$.)

One can show that when

$$-z \frac{U''(z)}{U'(z)} < 1,$$

$f'(z) > 0$, which implies that new optimistic information on future dividends leads to increased asset prices. In this case $p'(z) > 0$. (Of course, one might also expect that this information will lead to an attempted consumption binge now, lowering asset prices.)

Observationally, $p'(z)$ is the change in the ratio of a comprehensive stock price index to the CPI, as real output varies.

Even in this simplified economy, the relationship of asset prices to real output is far from simple and possibly not even monotonic. This is in line with Lucas' critique:

a macroeconomic model should be built from microeconomic principles.

Equity Premium Puzzle

Based on Mehra and Prescott (1985). In a model of the US economy with a representative agent, z corresponds to per-capita consumption. Using time series data on consumption, we can construct a Markov transition matrix.

The authors assume that the utility function is CRRA:

$$U(c) = \frac{c^{1-\sigma} - 1}{1-\sigma},$$

where σ is between one and three, based on micro observations.

Using our model, we can price assets. Suppose the price of a risk-free bond at state z is q_z^f . The (gross) return on the risk-free asset is

$$R_z^f = 1 + r_z^f = \frac{1}{q_z^f}.$$

The expected return for equity is

$$E_t[R_{t+1}^i|z] = \frac{\int [z'_i + p_i(z')]Q(z, dz')}{p_i(z)}.$$

The difference between the return on the equity and the return on the risk-free bond, $E_t(R_{t+1}|z) - R_z^f$, is the *risk premium* that the consumer needs to be paid in order to hold the risky asset.

Mehra and Prescott (1985) found that the maximum equity premium (based on the model) is 0.35% but the average equity premium in the stock market data (1889-1978) is 6.18%. This observation is what they called the *Equity Premium Puzzle*. To resolve it, $\sigma \approx 50$ is needed. When people are very risk averse, they need to be paid a lot to hold a risky asset. However, this value of σ is inconsistent with micro observations.

References

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MEHRA, R. AND E. C. PRESCOTT (1985): “The equity premium: A puzzle,” *Journal of Monetary Economics*, 15, 145–161.